

On the other hand, the fundamental group of a wedge sum of countably many circles is countably generated, hence countable.

The group $\pi_1(X)$ is actually far more complicated than $\prod_{\infty} \mathbb{Z}$. For one thing, it is nonabelian, since the retraction $X \rightarrow C_1 \cup \cdots \cup C_n$ that collapses all the circles smaller than C_n to the basepoint induces a surjection from $\pi_1(X)$ to a free group on n generators. For a complete description of $\pi_1(X)$ see [Cannon & Conner 2000].

It is a theorem of [Shelah 1988] that for a path-connected, locally path-connected compact metric space X , $\pi_1(X)$ is either finitely generated or uncountable.

Applications to Cell Complexes

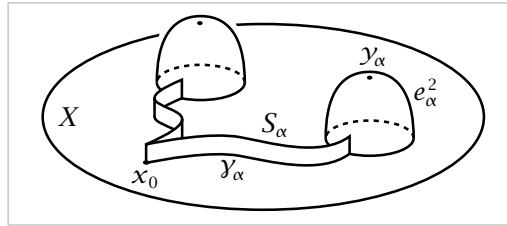
For the remainder of this section we shall be interested in 2-dimensional cell complexes, analyzing how the fundamental group is affected by attaching 2-cells. According to an exercise at the end of this section, attaching cells of higher dimension has no effect on π_1 , so all the interest lies in how the 2-cells are attached.

Suppose we attach a collection of 2-cells e_{α}^2 to a path-connected space X via maps $\varphi_{\alpha}: S^1 \rightarrow X$, producing a space Y . If s_0 is a basepoint of S^1 then φ_{α} determines a loop at $\varphi_{\alpha}(s_0)$ that we shall call φ_{α} , even though technically loops are maps $I \rightarrow X$ rather than $S^1 \rightarrow X$. For different α 's the basepoints $\varphi_{\alpha}(s_0)$ of these loops φ_{α} may not all coincide. To remedy this, choose a basepoint $x_0 \in X$ and a path γ_{α} in X from x_0 to $\varphi_{\alpha}(s_0)$ for each α . Then $\gamma_{\alpha}\varphi_{\alpha}\bar{\gamma}_{\alpha}$ is a loop at x_0 . This loop may not be nullhomotopic in X , but it will certainly be nullhomotopic after the cell e_{α}^2 is attached. Thus the normal subgroup $N \subset \pi_1(X, x_0)$ generated by all the loops $\gamma_{\alpha}\varphi_{\alpha}\bar{\gamma}_{\alpha}$ for varying α lies in the kernel of the map $\pi_1(X, x_0) \rightarrow \pi_1(Y, x_0)$ induced by the inclusion $X \hookrightarrow Y$.

Proposition 1.26. *The inclusion $X \hookrightarrow Y$ induces a surjection $\pi_1(X, x_0) \rightarrow \pi_1(Y, x_0)$ whose kernel is N . Thus $\pi_1(Y) \approx \pi_1(X)/N$.*

It follows that N is independent of the choice of the paths γ_{α} , but this can also be seen directly: If we replace γ_{α} by another path η_{α} having the same endpoints, then $\gamma_{\alpha}\varphi_{\alpha}\bar{\gamma}_{\alpha}$ changes to $\eta_{\alpha}\varphi_{\alpha}\bar{\eta}_{\alpha} = (\eta_{\alpha}\bar{\gamma}_{\alpha})\gamma_{\alpha}\varphi_{\alpha}\bar{\gamma}_{\alpha}(\gamma_{\alpha}\bar{\eta}_{\alpha})$, so $\gamma_{\alpha}\varphi_{\alpha}\bar{\gamma}_{\alpha}$ and $\eta_{\alpha}\varphi_{\alpha}\bar{\eta}_{\alpha}$ define conjugate elements of $\pi_1(X, x_0)$.

Proof: Let us expand Y to a slightly larger space Z that deformation retracts onto Y and is more convenient for applying van Kampen's theorem. The space Z is obtained from Y by attaching rectangular strips $S_{\alpha} = I \times I$, with the lower edge $I \times \{0\}$ attached along γ_{α} , the right edge $\{1\} \times I$ attached along an arc in e_{α}^2 , and all the left edges $\{0\} \times I$ of the different strips identified together. The top edges of the strips are not attached to anything, and this allows us to deformation retract Z onto Y .



In each cell e_α^2 choose a point y_α not in the arc along which S_α is attached. Let $A = Z - \bigcup_\alpha \{y_\alpha\}$ and let $B = Z - X$. Then A deformation retracts onto X , and B is contractible. Since $\pi_1(B) = 0$, van Kampen's theorem applied to the cover $\{A, B\}$ says that $\pi_1(Z)$ is isomorphic to the quotient of $\pi_1(A)$ by the normal subgroup generated by the image of the map $\pi_1(A \cap B) \rightarrow \pi_1(A)$. So it remains only to see that $\pi_1(A \cap B)$ is generated by the loops $y_\alpha \varphi_\alpha \bar{y}_\alpha$, or rather by loops in $A \cap B$ homotopic to these loops. This can be shown by another application of van Kampen's theorem, this time to the cover of $A \cap B$ by the open sets $A_\alpha = A \cap B - \bigcup_{\beta \neq \alpha} e_\beta^2$. Since A_α deformation retracts onto a circle in $e_\alpha^2 - \{y_\alpha\}$, we have $\pi_1(A_\alpha) \approx \mathbb{Z}$ generated by a loop homotopic to $y_\alpha \varphi_\alpha \bar{y}_\alpha$, and the result follows. $\square$

As a first application we compute the fundamental group of the orientable surface M_g of genus g . This has a cell structure with one 0-cell, $2g$ 1-cells, and one 2-cell, as we saw in Chapter 0. The 1-skeleton is a wedge sum of $2g$ circles, with fundamental group free on $2g$ generators. The 2-cell is attached along the loop given by the product of the commutators of these generators, say $[a_1, b_1] \cdots [a_g, b_g]$. Therefore

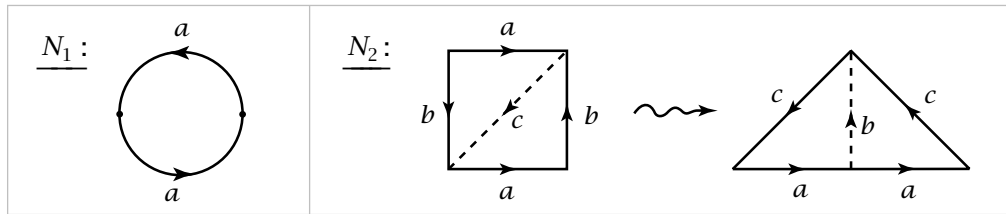
$$\pi_1(M_g) \approx \langle a_1, b_1, \dots, a_g, b_g \mid [a_1, b_1] \cdots [a_g, b_g] \rangle$$

where $\langle g_\alpha \mid r_\beta \rangle$ denotes the group with generators g_α and relators r_β , in other words, the free group on the generators g_α modulo the normal subgroup generated by the words r_β in these generators.

Corollary 1.27. *The surface M_g is not homeomorphic, or even homotopy equivalent, to M_h if $g \neq h$.*

Proof: The abelianization of $\pi_1(M_g)$ is the direct sum of $2g$ copies of $\mathbb{Z}$. So if $M_g \simeq M_h$ then $\pi_1(M_g) \approx \pi_1(M_h)$, hence the abelianizations of these groups are isomorphic, which implies $g = h$. $\square$

Nonorientable surfaces can be treated in the same way. If we attach a 2-cell to the wedge sum of g circles by the word $a_1^2 \cdots a_g^2$ we obtain a nonorientable surface N_g . For example, N_1 is the projective plane $\mathbb{R}P^2$, the quotient of D^2 with antipodal points of ∂D^2 identified. And N_2 is the Klein bottle, though the more usual representation



of the Klein bottle is as a square with opposite sides identified via the word $aba^{-1}b$. If one cuts the square along a diagonal and reassembles the resulting two triangles as shown in the figure, one obtains the other representation as a square with sides

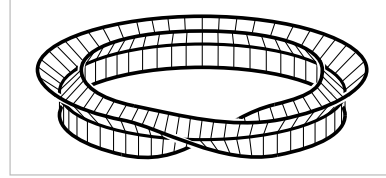
identified via the word a^2c^2 . By the proposition, $\pi_1(N_g) \approx \langle a_1, \dots, a_g \mid a_1^2 \cdots a_g^2 \rangle$. This abelianizes to the direct sum of $\mathbb{Z}_2$ with $g - 1$ copies of $\mathbb{Z}$ since in the abelianization we can rechoose the generators to be $a_1, \dots, a_{g-1}$ and $a_1 + \cdots + a_g$, with $2(a_1 + \cdots + a_g) = 0$. Hence N_g is not homotopy equivalent to N_h if $g \neq h$, nor is N_g homotopy equivalent to any orientable surface M_h .

Here is another application of the preceding proposition:

Corollary 1.28. *For every group G there is a 2-dimensional cell complex X_G with $\pi_1(X_G) \approx G$.*

Proof: Choose a presentation $G = \langle g_\alpha \mid r_\beta \rangle$. This exists since every group is a quotient of a free group, so the g_α 's can be taken to be the generators of this free group with the r_β 's generators of the kernel of the map from the free group to G . Now construct X_G from $\bigvee_\alpha S_\alpha^1$ by attaching 2-cells e_β^2 by the loops specified by the words r_β . $\square$

Example 1.29. If $G = \langle a \mid a^n \rangle = \mathbb{Z}_n$ then X_G is S^1 with a cell e^2 attached by the map $z \mapsto z^n$, thinking of S^1 as the unit circle in $\mathbb{C}$. When $n = 2$ we get $X_G = \mathbb{RP}^2$, but for $n > 2$ the space X_G is not a surface since there are n 'sheets' of e^2 attached at each point of the circle $S^1 \subset X_G$. For example, when $n = 3$ one can construct a neighborhood N of S^1 in X_G by taking the product of the graph Υ with the interval I , and then identifying the two ends of this product via a one-third twist as shown in the figure. The boundary of N consists of a single circle, formed by the three endpoints of each Υ cross section of N . To complete the construction of X_G from N one attaches a disk along the boundary circle of N . This cannot be done in $\mathbb{R}^3$, though it can in $\mathbb{R}^4$. For $n = 4$ one would use the graph $\times$ instead of Υ , with a one-quarter twist instead of a one-third twist. For larger n one would use an n -pointed 'asterisk' and a $1/n$ twist.



Exercises

1. Show that the free product $G * H$ of nontrivial groups G and H has trivial center, and that the only elements of $G * H$ of finite order are the conjugates of finite-order elements of G and H .
2. Let $X \subset \mathbb{R}^m$ be the union of convex open sets $X_1, \dots, X_n$ such that $X_i \cap X_j \cap X_k \neq \emptyset$ for all i, j, k . Show that X is simply-connected.
3. Show that the complement of a finite set of points in $\mathbb{R}^n$ is simply-connected if $n \geq 3$.